

Functional Network Connectivity Differences Associated with Depression and Lifetime Cannabis Use

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INTRODUCTION

- Depression is associated with altered functional network connectivity (FNC), although reported network-level patterns remain inconsistent.
- Cannabis use may influence systems involved in mood, cognition, reward, and emotion, but its relationship with depression-related connectivity remains unclear.

Objective: To examine the independent and interactive associations of depression diagnosis and cannabis use with resting-state FNC in UK Biobank participants.

METHODS

- Participants:** UK Biobank participants with resting-state fMRI scans, depression diagnosis, lifetime cannabis-use, demographic, and covariate data were included (**N = 4,007**).
- Diagnostic groups:** Participants with depression (**n = 838**; **11+ lifetime cannabis uses, n = 91**) were compared with controls (**n = 3,169**; **11+ lifetime cannabis uses, n = 291**). Controls with other reported psychiatric or neurological diagnoses were excluded.

Functional connectivity: Resting-state fMRI scans were analyzed using the NeuroMark 2.2 framework, which identified 105 intrinsic connectivity networks, producing 5,460 pairwise FNC connections for each participant.

Statistical analysis: General linear models tested the main effects of depression and cannabis use and the depression $\times$ cannabis interaction while adjusting for age, sex, ethnicity, mean framewise displacement, and imaging site. Multiple comparisons were evaluated using local FDR.

RESULTS

Figure 1 | Depression (a) and cannabis use (b) showed distinct but partially overlapping FNC patterns across cerebellar, paralimbic, subcortical/basal-ganglia, and default-mode domains. The depression $\times$ cannabis interaction (c) was more localized, particularly within CB-SA and SC/BG-SA connections. For the main-effect maps, red indicates higher FNC and blue indicates lower FNC in the depression group relative to controls and in the 11+ cannabis-use group relative to the no-use group. Gray cells in the upper triangle indicate connections that did not survive local FDR correction, and green brackets highlight representative domain-level patterns shown in Figure 2.

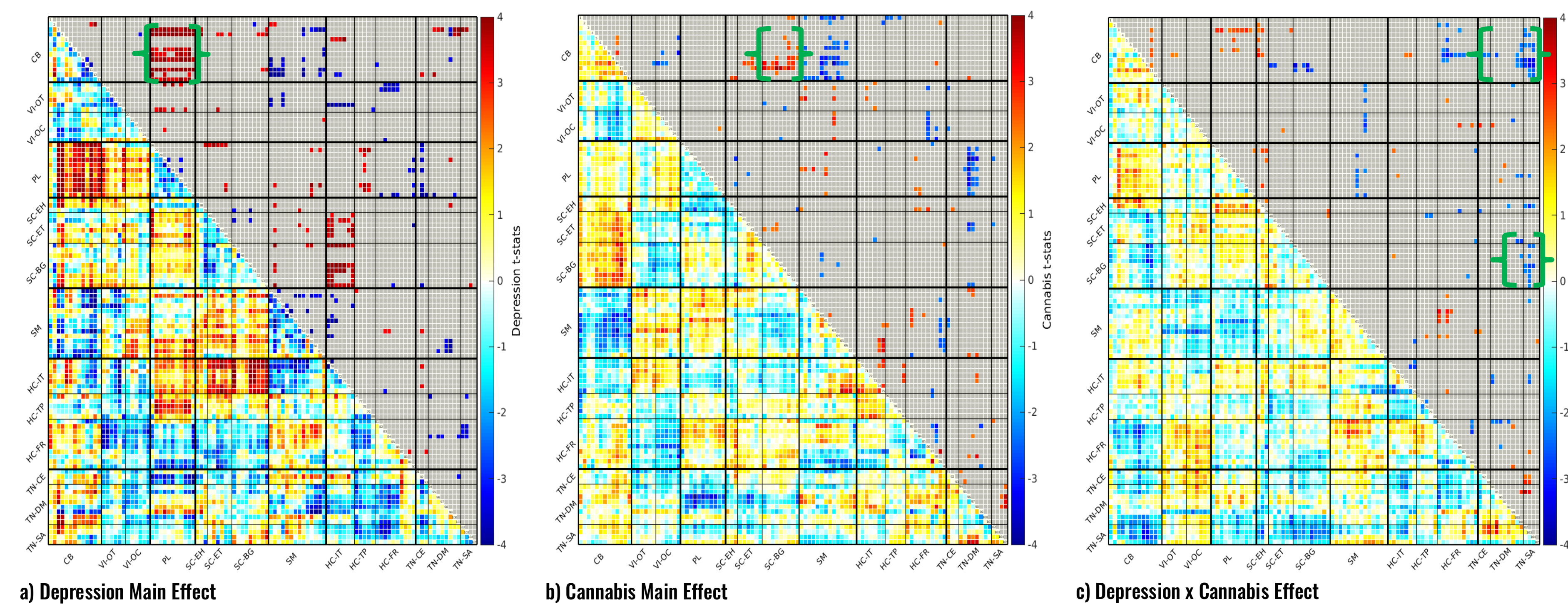


Figure 2 | Representative intrinsic connectivity network pairs selected from the highlighted domain-level patterns in Figure 1. Pairs were chosen to illustrate prominent effects within the CB-PL, CB-SC/BG, and SC/BG-SA blocks. Depression was associated with higher CB-PL FNC, cannabis use with higher CB-SC/BG FNC, and the depression $\times$ cannabis interaction involved SC/BG-SA connectivity.

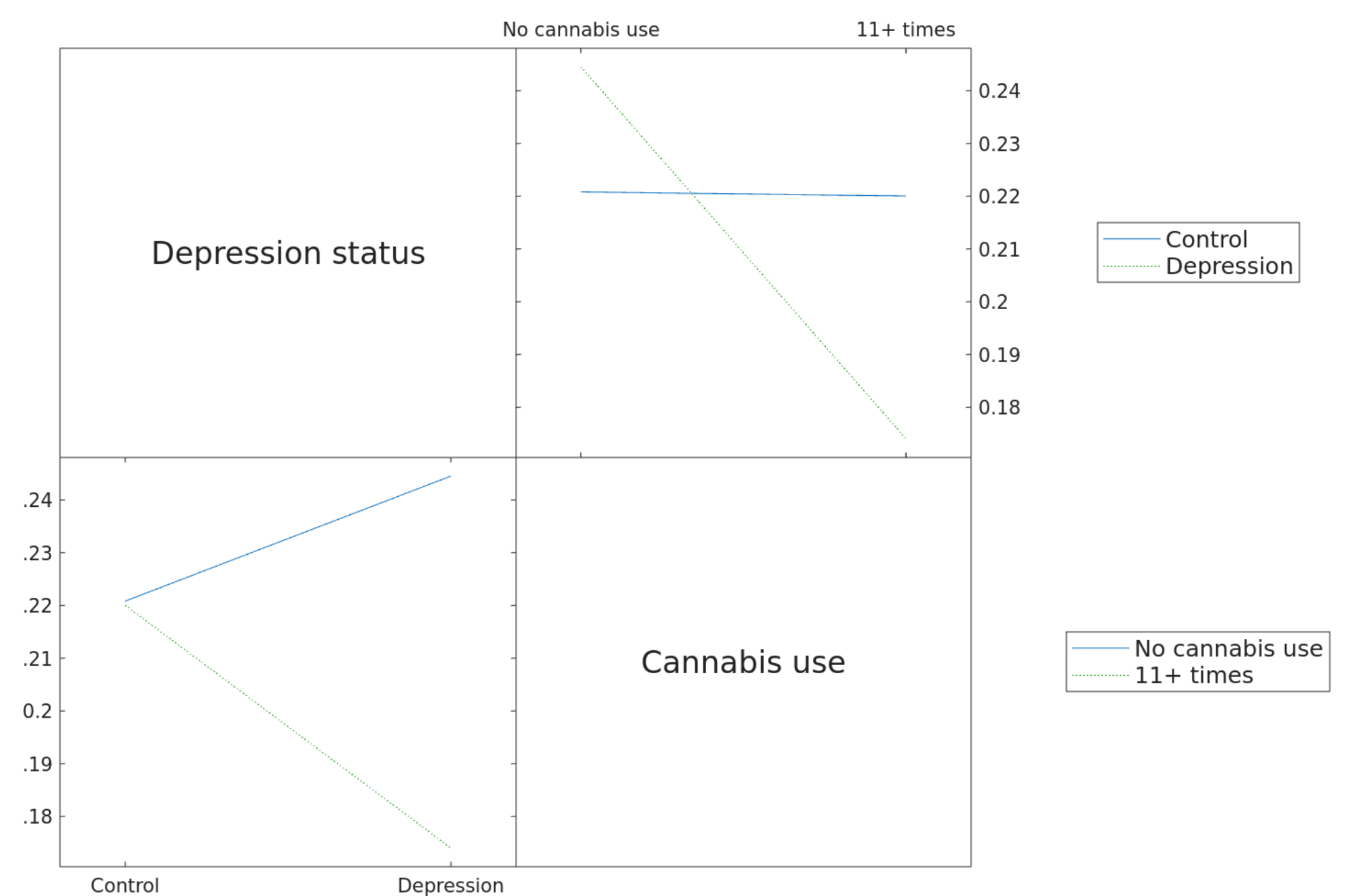
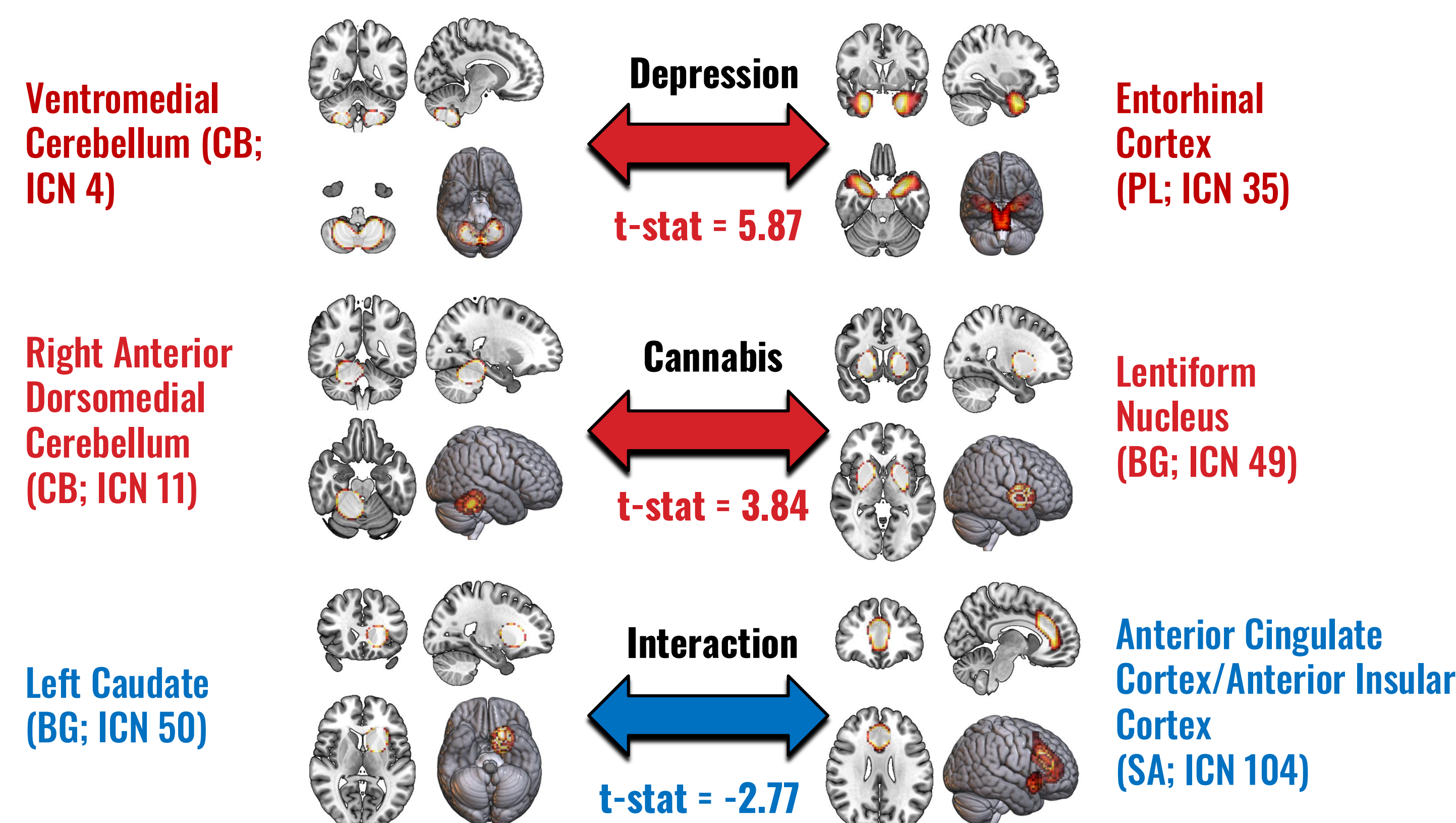


Figure 3 | Representative interaction plot for the SC/BG-SA connection (ICN 50–104), illustrating the depression $\times$ cannabis interaction identified in Figure 1. Controls showed relatively stable FNC across cannabis-use groups, whereas participants with depression showed markedly lower FNC in the 11+ lifetime-use group. Similar interaction patterns occurred across other SC/BG-SA and CB-SA connections. Participants reporting no lifetime cannabis use were compared with those reporting 11+ lifetime uses; intermediate-use categories were excluded.

DISCUSSION

- Key Findings:** Depression and cannabis use showed distinct but overlapping FNC patterns involving cerebellar, paralimbic, subcortical/basal-ganglia, and salience-related networks.
- Interpretation:** Because SC/BG components overlap with striatal regions such as the caudate and putamen, these findings may reflect altered coordination between reward and motivation systems and networks that identify behaviorally relevant information.
- Interaction Effects:** The association between lifetime cannabis use and salience-network FNC differed by depression status: controls with 11+ lifetime uses showed stable or higher SC/BG-SA and CB-SA FNC, whereas participants with depression showed lower FNC. This crossover pattern suggests that combined depression and cannabis-use history is associated with a distinct connectivity profile.
- Future Directions:** Future studies should examine cannabis recency, frequency, dose, and age of first use and determine whether depressive symptom severity, anhedonia, or dopamine-related reward processes contribute to these connectivity differences.

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